10Gbps RJ45

LabActiv-SFP
10Gbps Copper SFP Transceiver

Overview

LabActiv-SFP 10G Copper SFP Transceiver is Copper Small Form pluggable (SFP) transceiver is based on SFP multi-sourcing agreement (MSA). It is compatible with the 10GBASE-T and 1000BASE-T standards as specified in IEEE Std 802.3.The 1000BASE-T physical layer IC(PHY) can be accessed via I2C,allowing access to LIMITED PHY settings and features.

Features and Benefits

- Up to 10Gbps bi-directional data links
- RJ45 Max 30m
- Fully metallic enclosure for low EMI
- Compact RJ-45 Connector assembly
- Hot-pluggable SFP footprint
- Low power dissipation
- Extended case temperature: 0°C to 70°C

Regulator Compliance

The transceivers are Class 1 Laser Products comply with FDA regulations. Meet Class 1 eye safety requirements of EN 60825 and the electrical safety requirements of EN 60950.

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+3.3V Volt Electrical Power Interface

Parameter	Symbol	Min.	Type	Max	Units	Notes/Conditions
Supply Current	Is			600	mA	1.2W max power over full range of voltage and tem-perature.see caution note below
Power Supply Voltage	Vcc	3.13	3.3	3.47	v	Referebced to GND
Maximum Voltage	Vmax			4	v	
Surge Current	Isurge			550	mA	Hot plug above steady state current,See caution note below

Caution: Power consumption and surge current are higher than the specified values in the SFP MSA

Low-Speed Signals

Parameter	Symbol	Min.	Max	Units	Notes/Conditions
SFP Output LOW	VOL	0	0.5	V	4.7k to 10k pull-up to host Vcc,measured at host side of connector
SFP Output HIGH	VOH	Host Vcc-0.5	Host Vcc+0.3	V	4.7k to 10k pull-up to host Vcc,measured at host side of connector
SFP Input LOW	VIL	0	0.8	V	4.7k to 10k pull-up to Vcc,measured at SFP side of connector
SFP Input HIGH	VIH	2	Vcc+0.3	V	4.7k to 10k pull-up to Vcc,measured at SFP side of connector

High-Speed Electrical Interface

High-speed Electrical Interface Transmission Line-SFP

Parameter	Symbol	Min.	Type	Max	Units	Notes/Conditions
Line Frequency	fL	10	125	1000	MHZ	5-level encoding,per IEEE 802.3
Tx Output Impedance	Zout,TX		100		Ohm	Differential,for all frequencies between 1MHz and 125MHz
Rx Input Impedance	Zin,RX		100		Ohm	Differential,for all frequencies between 1MHz and 125MHz

High-speed Electrical Interface,Host-SFP

Parameter	Symbol	Min.	Type	Max	Units	Notes/Conditions
Single ended data input swing	Vinsing	180		700	mV	Single ended
Single ended data output swing	Voutsing	350		850	mV	Single ended
Rise/Fall Time	Tr,Tf		175		psec	20%-80%
Tx Input Impedance	Zin		50		Ohm	Single ended
Rx Output Impedance	Zout		50		Ohm	Single ended

General Specifications

Parameter	Symbol	Min.	Type	Max	Units	Notes/Conditions
Data Rate	BR	10		10000	Mb/sec	IEEE 802.3 compatible.See Notes 2 through 4 below
Tx Output Impedance	L			100	m	Category 5 UTP.BER<10^{-12}

Note:
1.Clock tolerance is +/- 50 ppm
2.By default,the is a full duplex device in preferred master mode
3.Automatic crossover detection is enabled. External crossover cable is not required
4. Multi- BASE-T operation requires the host system to have an SGMII interface with no clocks, and the module PHY to be configured per Applications Note AN-2036.With a SERDES that does not support SGMII,the module will operate at single rate only.

Environmental Specifications

The SFP has an extended range from 0°C to +70°C case temperature as specified in Table.

Parameter	Symbol	Min.	Type	Max	Units	Notes/Conditions
Operating Temperature	Top	0		70		Case temperature
Storage Temperature	Tsto	0		70		Ambient temperature

Order Information	
Part Number	**Description**
KA-CP10G-RJ45	LabActiv-SFP 10Gbps Copper SFP Transceiver